DO NOW — RECALL

Lesson 111 · OS Types, Boot Firmware, Drivers and VMs

Name: _____ Class: _____ Date: _____

1. In round robin, what happens when a running process's quantum expires but it is unfinished?

2. After pushing A then B onto a stack, which is popped first?

3. Why must a binary search operate on ordered data?

4. An OR gate has inputs 0 and 1. What is the output?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It returns to the ready queue (subject to scheduler rules).

2. Model answer

B, because a stack is LIFO.

3. Model answer

It discards half the search space using an ordering comparison.

4. Model answer

1.

By the end, you can…

• Embedded/real-time emphasise constraints; multi-user and distributed coordinate shared work.

• BIOS/UEFI firmware begins startup; drivers provide hardware-specific access.

• A hypervisor runs guest OSs; a language VM executes intermediate code.

Vocabulary — know these terms

real-time OS — System where meeting time constraints is essential.

BIOS/UEFI — Platform firmware that begins boot preparation.

hypervisor — Manages virtual machines running guest OSs.

device driver — Hardware-specific interface supplied to an OS.

NEW LEARNING

Read, annotate and explain

Match OS to use

Embedded/real-time emphasise constraints; multi-user and distributed coordinate shared work.

Boot and control

BIOS/UEFI firmware begins startup; drivers provide hardware-specific access.

Virtualise

A hypervisor runs guest OSs; a language VM executes intermediate code.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

An industrial safety controller must respond within a guaranteed deadline. Choose an OS type and justify it.

Model answer

A hard real-time operating system is suited because missing a deadline is unacceptable; deterministic scheduling/latency matters more than high average throughput. An embedded implementation may also apply.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain two differences between multi-user and multitasking operating systems, giving examples. [6]

Question 2

Describe the roles of BIOS/UEFI firmware and a device driver during startup and use. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare a hypervisor virtual machine with a bytecode virtual machine and explain a real-time deadline. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Every virtual machine must run a complete guest operating system.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.